UNIT - IV | DEEP LEARNING

CNNs, AlexNet, RNN, LSTM, GRU, RBM and MCMC

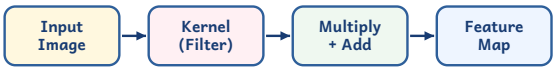
Q1. Define the 2D Convolution operation in CNNs.

Ans: 2D Convolution is an operation used in CNNs to find important features from an image.

1. A small matrix called a Kernel or Filter moves over the input image.
2. It multiplies the image values with the kernel values.
3. Then it adds the multiplied values to get a new value.
4. The final result is called a Feature Map.

Example: It helps to detect edges, shapes and patterns in images.

Simple sketch:



Q2. Give two key structural and functional differences between LeNet-5 and AlexNet.

Ans: LeNet-5 and AlexNet are two CNN architectures used for image recognition.

LeNet-5	AlexNet
Developed by Yann LeCun.	Developed by Alex Krizhevsky and his team.
Introduced in 1998.	Introduced in 2012.
Has a simple CNN architecture.	Has a more complex CNN architecture.
Uses Sigmoid or Tanh activation.	Mainly uses ReLU activation.
Uses grayscale images.	Uses RGB colour images.
Has fewer parameters.	Has more parameters.

Easy memory: LeNet = Simple CNN. AlexNet = Advanced CNN.

Q3. What is Pooling in CNNs? Differentiate between Max Pooling and Average Pooling.

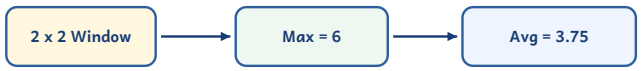
Ans: Pooling is a technique used in CNNs to reduce the size of feature maps.

1. It reduces the size of the image features.
2. It reduces computation and processing time.
3. It helps to keep important information.

Max Pooling	Average Pooling
Selects the maximum value from a window.	Calculates the average value from a window.
Keeps the highest value.	Keeps the average value.

Example: Input values: 1, 3, 5, 6. Max Pooling = 6. Average Pooling = (1+3+5+6)/4 = 3.75.

Simple sketch:



Q4. Define RNN and state why feedforward networks fail on sequential data.

Ans: RNN (Recurrent Neural Network) is a type of neural network used to process sequential data.

1. It remembers previous information using a hidden state.
2. It uses previous information to understand the current input.
3. It processes data step by step.
4. It is mainly used for text, speech and time-series data.

Why Feedforward Networks Fail:
1. They do not remember previous inputs.
2. They cannot easily understand the order of sequential data.

Example: RNN is useful for understanding sentences because word order is important.

Simple sketch:



Q5. What is the difference between Unidirectional RNN and Bidirectional RNN?

Ans: Both Unidirectional RNN and Bidirectional RNN are used to process sequential data. But they process information in different directions.

Unidirectional RNN	Bidirectional RNN
Processes data in one direction.	Processes data in two directions.
Reads from past to future.	Reads from past to future and future to past.
Uses previous information only.	Uses previous and future information.

Easy memory: Uni = One direction. Bi = Two directions.

Simple sketch:



Q6. Define LSTM network and name its three gates.

Ans: LSTM (Long Short-Term Memory) is a special type of RNN designed to learn long-term information from sequential data.

It uses a cell state and three main gates to control the flow of information.

1. Forget Gate: Decides which old information should be removed.
2. Input Gate: Decides which new information should be stored.
3. Output Gate: Decides which information should be given as output.

Easy memory: Forget = Remove. Input = Store. Output = Produce.

Simple sketch:



Q7. Differentiate between GRU and LSTM.

Ans: GRU and LSTM are special types of RNN used to learn information from sequential data.

GRU	LSTM
Has 2 gates.	Has 3 gates.
Uses Reset and Update gates.	Uses Forget, Input and Output gates.
Has a simple structure.	Has a more complex structure.
Requires less computation.	Requires more computation.
Does not have a separate cell state.	Uses a separate cell state.

Easy memory: GRU = 2 gates. LSTM = 3 gates.

Simple sketch:



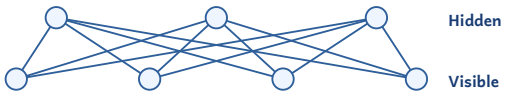
Q8. What is an RBM? Why is it called 'restricted'?

Ans: RBM (Restricted Boltzmann Machine) is a type of neural network used to learn important patterns and features from data.

It contains two main layers:
1. Visible Layer: It receives the input data.
2. Hidden Layer: It learns important features and patterns from the input data.

Why is it called restricted?
It is called restricted because neurons in the same layer are not connected to each other. Connections exist only between the visible layer and hidden layer.

Simple sketch:



Q9. Define MCMC and Gibbs Sampling in generative energy models.

Ans: MCMC (Markov Chain Monte Carlo) is a sampling method used to generate samples from complex probability distributions.

Gibbs Sampling:
1. Gibbs Sampling is a type of MCMC method.
2. It changes one variable at a time while keeping other variables fixed.
3. It is used in Restricted Boltzmann Machines (RBMs) to generate samples.
4. It helps the model learn patterns from data.

Easy memory: MCMC = Sampling method. Gibbs Sampling = Changes one variable at a time.

Simple sketch:

